Matched-equilibrium lower bound for polynomial current noise in a torque-driven underdamped Gaussian particle

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An equilibrium process with the same position marginal as a nonequilibrium underdamped diffusion provides a natural reference for current fluctuations, but the reference need not order arbitrary observables. We resolve this comparison for a stable two-dimensional Ornstein–Uhlenbeck particle driven by a constant torque and for the complete family of position-current weights whose homogeneous degrees are one and three. Gradient weights are stationary boundary increments, leaving a four-dimensional quotient of the original twelve-dimensional coefficient space. After an exact nondimensionalization, rotational covariance splits that quotient into a coupled isotropic block and an angular-two doublet. Complete polynomial Poisson equations of total degree at most four give the driven-minus-equilibrium diffusion matrix as rational functions of two positive parameters. Its isotropic leading minor and determinant and its angular coefficient have explicit strictly positive certificates over the full stability domain. Consequently, every current in the stated family is at least as noisy in the driven process as in the matched equilibrium trap, with equality exactly for gradient boundary currents. The calculation is exact for the linear dynamics and is independently checked by a two-time covariance and Wick-contraction representation.

I. INTRODUCTION

Current fluctuations quantify the precision of transport in nonequilibrium steady states. In underdamped systems their behavior is especially sensitive to the observable and to forces that act on velocity or break time-reversal symmetry. For example, a magnetic field can invalidate familiar thermodynamic-uncertainty bounds for work and heat currents [1]. It is therefore useful to separate statements about a particular current from statements that order a whole family of observables.

Dechant introduced an "equivalent equilibrium" dynamics for an underdamped steady state: the reference retains the position marginal while removing the local mean flow. In a torque-driven Brownian-gyrator sector, the torque-work current is noisier than its equivalent-equilibrium counterpart, and the same ordering was conjectured for more general observables [2]. The conjecture cannot hold without specifying both the model and the observable class. Recent velocity-resolved examples show that closely related fluctuation inequalities can fail when the readout or physical premises change [3].

Here we give an all-parameter result for a finite but complete polynomial sector of the zero-field torque-driven Gaussian model. The weight of the position current may contain every real homogeneous term of degree one and three. This sector is already beyond the linear current-type observables in the exact large-deviation treatment of linear diffusions by du Buisson and Touchette [4]; cubic position weights produce quartic phase-space rates. Their erratum concerns eigenfunction normalizability and full scaled-cumulant-generating-function domains [5], whereas our argument uses only the centered Poisson equation at zero tilt. Results for occupation estimators in nonreversible samplers also have different matching premises: those processes share the full invariant law, not only the position marginal [6].

Our result has three ingredients. First, quotienting gradient boundary increments reduces the twelve current coefficients to four without discarding any extensive current. Second, a change of units leaves only two positive dimensionless parameters and preserves the sign of the diffusion difference by a nonsingular congruence. Third, rotational covariance reduces the exact degree-four Poisson calculation to a two-dimensional invariant block and an angular-two doublet. The resulting global sign certificate proves the ordering throughout the strict stability domain, rather than at sampled parameters. The closest sources identified above do not state this cubic quotient theorem; that comparison is necessarily a bounded priority search, not a claim of exhaustive bibliographic precedence.

II. MODEL, STATIONARY LAWS, AND CURRENT SPACE

Let

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad x, v \in \mathbb{R}^2. \tag{1}$$

The driven process is

$$\mathrm{d}x_t = v_t \,\mathrm{d}t, \tag{2a}$$

$$m\,\mathrm{d}v_t = (-kx_t + \kappa R x_t - \gamma v_t)\,\mathrm{d}t + \sqrt{2\gamma T}\,\mathrm{d}W_t, \tag{2b}$$

where W_t is standard two-dimensional Brownian motion. We assume

$$m, \gamma, k, T > 0, \qquad \kappa \neq 0, \qquad k_* := k - \frac{m\kappa^2}{\gamma^2} > 0. \tag{3}$$

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The comparison process keeps m, γ, T but replaces the entire position force with the conservative force $-k_* x$:

$$\mathrm{d}x_t = v_t\,\mathrm{d}t, \tag{4a}$$

$$m\,\mathrm{d}v_t = (-k_* x_t - \gamma v_t)\,\mathrm{d}t + \sqrt{2\gamma T}\,\mathrm{d}W_t. \tag{4b}$$

The matched stiffness is k_*, not k.

Both processes are initialized in their own complete stationary laws. Direct solution of the Lyapunov equation gives, for Eq. (2),

$$x \sim N\!\left(0, \frac{T}{k_*} I_2\right), \qquad v \mid x \sim N\!\left(\frac{\kappa}{\gamma} R x, \frac{T}{m} I_2\right), \tag{5}$$

whereas Eq. (4) has

$$x \sim N\!\left(0, \frac{T}{k_*} I_2\right), \qquad v \sim N\!\left(0, \frac{T}{m} I_2\right), \qquad x \perp v. \tag{6}$$

Thus the position marginals agree although the phase-space laws do not. For each complex spatial mode the deterministic characteristic equation is $m\lambda^2 + \gamma\lambda + k \mp i\kappa = 0$; its strict Hurwitz condition is $m\kappa^2 < \gamma^2 k$, exactly Eq. (3). There is no spatial cutoff or boundary other than Gaussian decay at infinity. We set Boltzmann's constant to one, so T has units of energy, while k and κ have units of stiffness.

Let $\mathcal{H}_d$ denote the real homogeneous scalar polynomials of degree d in (x_1, x_2), and define the twelve-dimensional current-weight space

$$\mathcal{W}_{1,3} = (\mathcal{H}_1 \oplus \mathcal{H}_3)^2. \tag{7}$$

For $w \in \mathcal{W}_{1,3}$, use the same weight in both processes and set

$$J_w(t) = \int_0^t w(x_s) \cdot v_s\,\mathrm{d}s, \qquad D(w) = \lim_{t\to\infty} \frac{\operatorname{Var} J_w(t)}{2t}. \tag{8}$$

The two diffusion rates will be denoted D_d and D_e. Because $\mathrm{d}x = v\,\mathrm{d}t$ has finite variation, Eq. (8) has no Itô–Stratonovich ambiguity. The variance is centered by definition; the two current means need not agree.

III. GRADIENT QUOTIENT AND STATEMENT OF THE RESULT

For any $\phi \in \mathcal{H}_2 \oplus \mathcal{H}_4$,

$$J_{\nabla\phi}(t) = \phi(x_t) - \phi(x_0). \tag{9}$$

Stationarity and Gaussian moments make its variance bounded uniformly in t. Its covariance with an extensive current is $o(t)$ by Cauchy–Schwarz. Hence both diffusion forms are invariant under $w \mapsto w + \nabla\phi$, and the relevant space is

$$\mathcal{W}_{1,3}/\mathcal{G}, \qquad \mathcal{G} = \nabla(\mathcal{H}_2 \oplus \mathcal{H}_4). \tag{10}$$

The gradient space has dimension $3 + 5 = 8$, leaving a four-dimensional quotient.

Every quotient class has the unique representative

$$w_c(x) = \left(c_0 + c_1 x_1^2 + c_2 x_1 x_2 + c_3 x_2^2\right) Rx. \tag{11}$$

Indeed, in two dimensions curl has the gradient fields as its kernel, while

$$\operatorname{curl}(pRx) = 2p + x \cdot \nabla p. \tag{12}$$

It acts as multiplication by 2 on constant p and by 4 on quadratic homogeneous p. Dimension counting then proves existence and uniqueness in Eq. (11); further details are in Appendix A.

Theorem 1 *For every parameter tuple satisfying Eq.* (3) *and every real* $w \in \mathcal{W}_{1,3}$,

$$D_\mathrm{d}(w) \geq D_\mathrm{e}(w). \tag{13}$$

Equality holds if and only if $w \in \nabla(\mathcal{H}_2 \oplus \mathcal{H}_4)$.

The theorem concerns only the stationary long-time second-cumulant rate. It does not assert a finite-time inequality, a full generating-function ordering, or an ordering for velocity-dependent or higher-degree weights.

IV. DIMENSIONLESS EXACT POISSON REPRESENTATION

Introduce dimensionless time and length

$$\theta = \frac{\gamma t}{m}, \qquad \ell = \frac{\sqrt{mT}}{\gamma}, \qquad X = \frac{x}{\ell}, \qquad V = \frac{mv}{\gamma\ell}, \tag{14}$$

and the two parameters

$$a = \frac{mk_*}{\gamma^2} > 0, \qquad b = \frac{m\kappa}{\gamma^2} \neq 0, \qquad u = b^2 > 0. \tag{15}$$

The backward generators become

$$L_b = V \cdot \nabla_X + [-(a + b^2)X + bRX - V] \cdot \nabla_V + \Delta_V, \tag{16a}$$

$$L_0 = V \cdot \nabla_X + [-aX - V] \cdot \nabla_V + \Delta_V. \tag{16b}$$

Their covariance blocks are

$$\begin{aligned} \langle XX^\mathsf{T}\rangle &= a^{-1} I, \\ \langle VX^\mathsf{T}\rangle &= (b/a) R, \\ \langle VV^\mathsf{T}\rangle &= ((a + b^2)/a) I \end{aligned} \tag{17}$$

for the driven process and $(a^{-1}I, 0, I)$ for equilibrium.

Let $S_\ell = \operatorname{diag}(1, \ell^2, \ell^2, \ell^2)$ act on the coefficient vector in Eq. (11). Direct substitution in the integrated current gives

$$D_\mathrm{physical}(c) = \frac{\gamma\ell^4}{m} D_\mathrm{dimensionless}(S_\ell c). \tag{18}$$

The prefactor is positive and S_ℓ is nonsingular. Thus sign and kernel dimension are preserved. Torque reversal is implemented by a spatial reflection; the final symmetric form depends on b only through $u = b^2$.

For a centered polynomial rate $\widehat{f} = f - \langle f \rangle$, solve

$$-Lh = \widehat{f}. \tag{19}$$

Exponential stability makes the centered solution unique up to a constant, and the Green–Kubo formula gives

$$D(f) = \langle \widehat{f} h \rangle. \tag{20}$$

For several basis currents the quadratic-form matrix is the symmetric part of $\langle \widehat{f}_i h_j \rangle$. Because L_b and L_0 map phase-space polynomials of total degree at most four into the same space, the calculation is an exact finite closure, not a truncation of the dynamics.

Writing

$$\mathcal{A} = X_1 V_2 - X_2 V_1, \tag{21}$$

the rates associated with Eq. (11) are a quadratic prefactor times $\mathcal{A}$. Rotational covariance separates the invariant span of $\mathcal{A}$ and $|X|^2\mathcal{A}$ from a real angular-two doublet. The complete invariant Poisson sector contains 13 nonconstant monomials and the complex angular-two sector contains 11. Appendix B lists these sectors and their stationary means.

V. EXACT QUOTIENT MATRIX AND POSITIVITY CERTIFICATE

Decompose the quadratic prefactor as

$$p = c_0 + s|X|^2 + d(X_1^2 - X_2^2) + eX_1X_2, \tag{22}$$

where

$$s = \frac{c_1 + c_3}{2}, \qquad d = \frac{c_1 - c_3}{2}, \qquad e = c_2. \tag{23}$$

The exact driven-minus-equilibrium calculation gives

$$\Delta D = Ac_0^2 + 2Bc_0 s + Cs^2 + Gd^2 + \frac{G}{4}e^2. \tag{24}$$

Define

$$A = \frac{2u(5a + 4u + 1)}{a^3}, \tag{25a}$$

$$B = \frac{16u(4a + 4u + 1)}{a^4}, \tag{25b}$$

$$H = 16a^2 + 32au + 24a + 16u^2 + 40u + 9, \tag{25c}$$

$$\Delta_0 = \frac{32u^2 P_0(a,u)}{a^8(4a+3)H}, \tag{25d}$$

$$C = \frac{\Delta_0 + B^2}{A}. \tag{25e}$$

The angular coefficient is

$$G = \frac{16uP_2(a,u)}{a^3(4a+3)(4a+3u+3)((a+u)^2+u)K}, \tag{26}$$

with

$$K = (16a^2+32au+12a+16u^2-13u)^2+4u(20a+20u+3)^2. \tag{27}$$

The integer polynomials P_0 and P_2 are displayed in Appendix C. All 20 coefficients of P_0 and all 33 coefficients of P_2 are strictly positive.

In the ordered quotient basis $(1, X_1^2, X_1X_2, X_2^2)RX$, Eq. (24) is the matrix

$$Q_\Delta = \begin{pmatrix} A & B/2 & 0 & B/2 \\ B/2 & (C+G)/4 & 0 & (C-G)/4 \\ 0 & 0 & G/4 & 0 \\ B/2 & (C-G)/4 & 0 & (C+G)/4 \end{pmatrix}. \tag{28}$$

Every factor in the denominators of Eqs. (25) and (26) is positive when $a, u > 0$. In particular, $K > 0$ follows immediately from Eq. (27). Positive polynomial coefficients imply $P_0, P_2 > 0$. Consequently

$$A > 0, \qquad \det \begin{pmatrix} A & B \\ B & C \end{pmatrix} = \Delta_0 > 0, \qquad G > 0. \tag{29}$$

Sylvester's criterion makes the isotropic block positive definite, while the angular-two doublet is strictly positive. Hence Q_Δ is positive definite on the four-dimensional quotient for every $a, u > 0$. The scaling in Eq. (18) transfers this statement to every physical tuple in Eq. (3). Gradient classes have zero long-time rate in both processes, and every nongradient class has $\Delta D > 0$. This proves Theorem 1, including its equality statement.

VI. BENCHMARK AND INDEPENDENT CHECKS

Before applying the closure to cubic currents, we reproduced the published zero-field torque-work benchmark. With $W(t) = \kappa \int_0^t (Rx_s) \cdot v_s \, \mathrm{d}s$, the source convention gives

$$\langle \dot{W} \rangle = \frac{2\kappa^2 T}{\gamma k_*}, \tag{30a}$$

$$D_W = \frac{2\kappa^2 T^2}{\gamma k_*^3} \left(k^2 + \kappa^2 + \frac{3mk\kappa^2}{\gamma^2} \right), \tag{30b}$$

in agreement with the zero-magnetic-field specialization of Ref. [1]. That reference uses the opposite torque orientation, $\kappa_{\mathrm{CFS}} = -\kappa$; Eq. (30) is even in torque. Independent real-coordinate Poisson and complex-coordinate two-time covariance reconstructions reproduce both formulas.

The global proof is symbolic. Four Poisson identities—two invariant and two angular, driven and equilibrium—have zero polynomial residual. The positive-coefficient

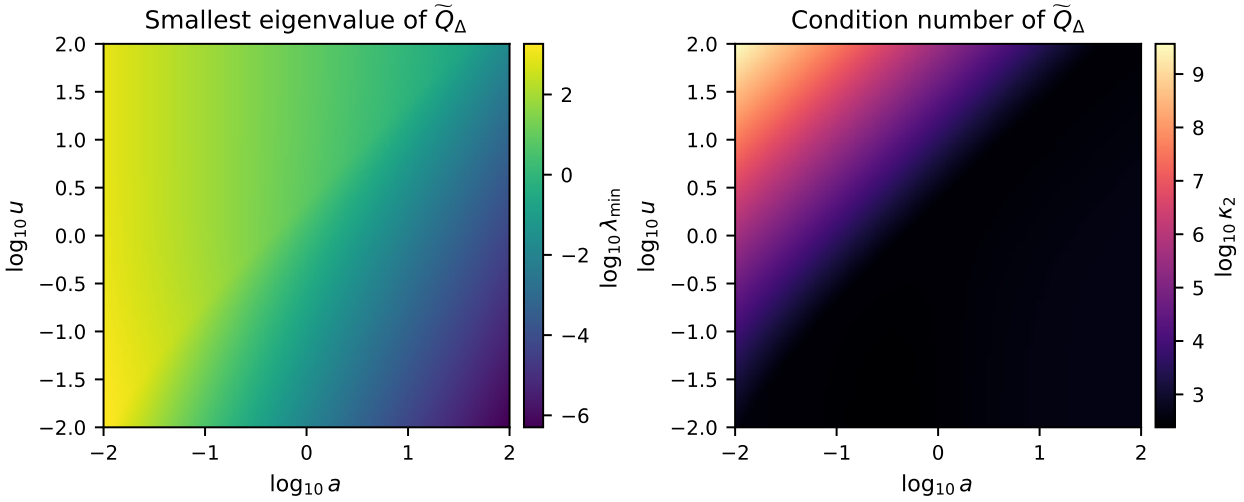


FIG. 1. Numerical visualization of the exact quotient formula on an 81×81 logarithmic grid with $10^{-2} \le a, u \le 10^2$. Left: smallest eigenvalue of the thermally normalized $\widetilde{Q}_\Delta$. Right: its spectral condition number. Every sampled eigenvalue is positive, but the grid is illustrative only; positivity on the continuous domain follows from Eqs. (25)– (29). The plotted values and regeneration script accompany the manuscript. OpenAI Codex assisted in writing and checking the deterministic plotting code; the plotted data were generated from the exact formulas, as detailed in Sec. VIII.

audit gives minimum integer coefficients 27 for P_0 and 1620 for P_2. Complete real polynomial solves with 69 nonconstant unknowns agree exactly with Eq. (28) at the rational points

$$(a, b) = (1, 1), \quad (2, 1/2), \quad (3/2, -2/3), \tag{31}$$

including all constant Poisson equations. These hold-outs test the reduction and torque sign but do not replace the continuous-domain certificate.

As a separate numerical representation, we form the full two-time covariance of (X, V), evaluate degree-eight Gaussian moments by Wick contractions, and integrate symmetrized correlations on $[0, \infty)$. Across the three physical cases listed in the reproducibility data, the largest Lyapunov residual is 5.77×10^{-14}, the largest reported quadrature error is 5.99×10^{-9}, and the largest Poisson–Wick relative matrix discrepancy is 1.053×10^{-12}. These numerical values are auxiliary error controls; the theorem itself rests on exact identities.

Figure 1 illustrates the thermally normalized quotient spectrum. To put linear and cubic coefficients on comparable stationary scales, write $c = (\widetilde{c}_0, a\widetilde{c}_1, a\widetilde{c}_2, a\widetilde{c}_3)$, so $\widetilde{Q}_\Delta = \mathrm{diag}(1, a, a, a) Q_\Delta \mathrm{diag}(1, a, a, a)$. The wide variation of its condition number also explains why a floating-point scan alone would be a poor global sign test.

VII. PHYSICAL INTERPRETATION AND LIMITATIONS

The equilibrium reference in Eq. (4) is matched only at the level of the position marginal. The driven phase-space law retains a rotational conditional velocity, Eq. (5). Theorem 1 states that, despite this distinction, eliminating the rotation cannot increase the long-time noise of any degree-one/three position current. The result includes anisotropic cubic readouts: none can undercut the matched equilibrium calibration. Equality has a concrete kinematic meaning, namely a nonextensive endpoint difference $\phi(x_t) - \phi(x_0)$, rather than a hidden finite-current saturation.

The conclusion is deliberately local in model and observable space. It uses linear Gaussian dynamics, zero magnetic field, stationary initialization, strict stability, and the second cumulant at long times. It does not establish the general-observable conjecture of Ref. [2]. It also does not apply at the stability boundary, to nonlinear forces, to velocity-dependent weights, to finite-time variances, to higher cumulants, or to full large-deviation domains. In particular, the integrability corrections in Ref. [5] caution against promoting a local Poisson calculation to an unproved statement about the complete scaled cumulant generating function.

The quotient construction and rotational-sector decomposition are reusable: they turn a broad coefficient search into exact low-dimensional sign tests. For weights of higher odd degree the OU polynomial closure remains finite, but new angular sectors and larger positivity problems appear. Whether the matched-equilibrium ordering persists there is left open.

VIII. COMPUTATIONAL REPRODUCIBILITY AND AI ASSISTANCE

The accompanying package contains the exact symbolic Poisson calculation, complete rational hold-outs,

the independent covariance/Wick implementation, the benchmark reconstruction, figure-generation code, environment versions, and reference outputs. All calculations are deterministic. The exact residual tests and positive-domain formulas, rather than numerical precision, are the basis of the result.

OpenAI Codex using the model identifier `gpt-5.6-sol` provided substantive assistance with scientific reasoning, symbolic and numerical code, literature synthesis, verification design, figure generation, and manuscript drafting. It was directed to retain the full parameter domain, matched stiffness, stationary boundary conditions, units, and factor-of-two convention. Its mathematical output was checked by exact zero-residual identities, positive-coefficient certificates, complete rational hold-outs, and a separately implemented covariance/Wick calculation as described above. These machine checks do not replace author judgment or accountability; the named author must complete a final scientific, textual, rights, and policy review before any submission.

DATA AVAILABILITY

No experimental data were used. All code and generated numerical data needed to reproduce the formulas, checks, and figure are included with this manuscript in the reproducibility package described in Sec. VIII. Downloaded third-party articles are not part of that package.

Appendix A: Completeness of the gradient quotient

For a homogeneous vector field of degree d in two variables, the vector space has dimension $2(d+1)$. Gradients of $\mathcal{H}_{d+1}$ form a subspace of dimension $d+2$: a homogeneous polynomial of positive degree cannot have identically zero gradient. The scalar curls have degree $d-1$ and span a space of dimension d. Since

$$2(d+1)-(d+2)=d, \tag{A1}$$

the curl map induces an isomorphism from vector fields modulo gradients to $\mathcal{H}_{d-1}$. Equation (12) supplies an explicit inverse on the complement pRx. Applying this argument for $d=1$ and $d=3$ gives the four representatives in Eq. (11) and proves that no extensive position-current direction is lost.

For completeness, if $g=\nabla\phi$ and f is any polynomial current rate, stationarity gives

$$\mathrm{Var}[\phi(x_t)-\phi(x_0)]=O(1), \qquad \mathrm{Var}\, J_f(t)=O(t). \tag{A2}$$

Therefore

$$\frac{|\,\mathrm{Cov}(J_f(t),\phi(x_t)-\phi(x_0))|}{t}\le\frac{O(\sqrt{t})}{t}\longrightarrow 0. \tag{A3}$$

This proves invariance of the entire asymptotic bilinear form, not merely its diagonal values.

Appendix B: Complete symmetry-reduced polynomial sectors

Set

$$\begin{aligned} X_0&=|X|^2, \quad V_0=|V|^2,\\ C_0&=X\cdot V, \quad \mathcal{A}=X_1V_2-X_2V_1. \end{aligned} \tag{B1}$$

A linearly independent basis of all nonconstant rotational invariants through total degree four is

$$\begin{aligned} X_0, V_0, C_0, \mathcal{A}, X_0^2, V_0^2, X_0V_0, X_0C_0, X_0\mathcal{A},\\ V_0C_0, V_0\mathcal{A}, C_0^2, C_0\mathcal{A}. \end{aligned} \tag{B2}$$

The identity $\mathcal{A}^2=X_0V_0-C_0^2$ removes the only redundant quartic. Both generators in Eq. (16) preserve this 13-dimensional sector. For the targets $(\mathcal{A}, X_0\mathcal{A})$, the driven stationary means are

$$\langle\mathcal{A}\rangle_b=\frac{2b}{a}, \qquad \langle X_0\mathcal{A}\rangle_b=\frac{8b}{a^2}, \tag{B3}$$

and both equilibrium means vanish.

For the angular-two sector introduce independent formal complex variables $z=X_1+iX_2$, $q=V_1+iV_2$ and their conjugates. The degree-two basis is (z^2, zq, q^2). The eight degree-four monomials are

$$z^jq^{3-j}\bar{z}, \quad z^jq^{3-j}\bar{q}, \qquad j=0,1,2,3. \tag{B4}$$

They are precisely the degree-two and degree-four monomials of rotational weight $+2$. The target $z^2\mathcal{A}$ lies in their span after application of the generator, and its stationary mean is zero. Direct substitution of each solved polynomial h in Eq. (19) gives an identically zero residual. Gaussian moments are evaluated recursively by pairings from the covariances in Eq. (17).

Appendix C: Positive polynomials

The isotropic determinant in Eq. (25) contains

$$\begin{aligned} P_0(a,u) = {} & 448a^5 + 2944a^4u + 1700a^4 + 5568a^3u^2 + 8220a^3u + 2683a^3 \\ & + 4096a^2u^3 + 12480a^2u^2 + 8432a^2u + 1872a^2 \\ & + 1024au^4 + 6144au^3 + 8256au^2 + 3424au + 432a \\ & + 768u^4 + 2304u^3 + 1440u^2 + 336u + 27. \end{aligned} \tag{C1}$$

Its 20 integer coefficients are positive. The angular coefficient in Eq. (26) contains

$$\begin{aligned} P_2(a,u) = {} & 10496a^7 + 60928a^6u + 41024a^6 + 147200a^5u^2 + 235488a^5u + 70512a^5 \\ & + 189440a^4u^3 + 529920a^4u^2 + 359932a^4u + 65772a^4 \\ & + 136960a^3u^4 + 591680a^3u^3 + 702445a^3u^2 + 278523a^3u + 28404a^3 \\ & + 52736a^2u^5 + 335040a^2u^4 + 633102a^2u^3 + 458033a^2u^2 + 105507a^2u + 3564a^2 \\ & + 8448au^6 + 83424au^5 + 251037au^4 + 299256au^3 + 142527au^2 + 19332au \\ & + 4608u^6 + 30960u^5 + 61749u^4 + 50670u^3 + 16893u^2 + 1620u. \end{aligned} \tag{C2}$$

All 33 coefficients are positive. Thus $P_0(a,u) > 0$ and $P_2(a,u) > 0$ for the entire open domain $a, u > 0$, with no numerical lower-bound argument.

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